

A Theta-Kernel Lift of Somos-4

Internal research note, version 0.1

Thomas Scheuerle

23 September 2026

Abstract

Let $(H_n)_{n \in \mathbb{Z}}$ be a bilateral Somos-4 sequence. Scaling a moment generating function by $G(x) \mapsto G(qx)$ multiplies its n th Hankel determinant by $q^{n(n-1)}$. This naturally suggests the bilateral theta-kernel transform

$$S(t, q) = \sum_{n \in \mathbb{Z}} H_n q^{n(n-1)} t^n.$$

We show that the Somos-4 recurrence becomes an exact relation between three fixed difference diagonals of the quadratic kernel

$$\mathcal{K}(t, z; q) = S(tz, q)S(t/z, q).$$

Specifically, the z -diagonals of separations 4, 2, and 0 satisfy a linear relation. This is a rigorous first bridge between the Hankel/Somos construction and a theta-like bilateral q -series, without introducing an elliptic parametrization.

1 Starting point

Assume that a bilateral sequence $(H_n)_{n \in \mathbb{Z}}$ satisfies

$$H_{n+4}H_n = \alpha H_{n+3}H_{n+1} + \beta H_{n+2}^2, \quad n \in \mathbb{Z}. \quad (1)$$

If

$$G(x) = \sum_{k \geq 0} g_k x^k, \quad H_n = \det(g_{i+j})_{0 \leq i, j < n},$$

then

$$G(qx) = \sum_{k \geq 0} q^k g_k x^k.$$

Factoring q^i from row i and q^j from column j gives

$$H_n[G(qx)] = q^{n(n-1)} H_n[G]. \quad (2)$$

Thus the quadratic exponent is forced by the elementary scaling of the moment variable.

2 The weighted Somos sequence

Define

$$W_n = H_n q^{n(n-1)}. \quad (3)$$

A direct comparison of exponents gives

$$\begin{aligned}(n+4)(n+3) + n(n-1) &= 2n^2 + 6n + 12, \\ (n+3)(n+2) + (n+1)n &= 2n^2 + 6n + 6, \\ 2(n+2)(n+1) &= 2n^2 + 6n + 4.\end{aligned}$$

Consequently, (1) is equivalent to

$$W_{n+4}W_n = \alpha q^6 W_{n+3}W_{n+1} + \beta q^8 W_{n+2}^2. \quad (4)$$

In particular, the scaling $G(x) \mapsto G(qx)$ induces the parameter transformation

$$(\alpha, \beta) \mapsto (\alpha q^6, \beta q^8).$$

3 The bilateral theta-kernel transform

Define the formal bilateral series

$$S(t, q) = \sum_{n \in \mathbb{Z}} W_n t^n = \sum_{n \in \mathbb{Z}} H_n q^{n(n-1)} t^n. \quad (5)$$

Since

$$q^{n(n-1)} t^n = q^{n^2} (t/q)^n,$$

S is a coefficientwise Somos deformation of the standard bilateral theta kernel $\sum_{n \in \mathbb{Z}} q^{n^2} u^n$. No claim is made here that S is itself a classical theta function.

Now introduce the quadratic kernel

$$\mathcal{K}(t, z; q) = S(tz, q)S(t/z, q). \quad (6)$$

Its expansion is

$$\mathcal{K}(t, z; q) = \sum_{i, j \in \mathbb{Z}} W_i W_j t^{i+j} z^{i-j}. \quad (7)$$

The exponent of t records the sum of the indices, while the exponent of z records their difference.

Proposition 1 (Diagonal form of Somos-4). *Let S and $\mathcal{K}$ be defined by (5) and (6). Then (4) is equivalent to*

$$[z^4]\mathcal{K}(t, z; q) = \alpha q^6 [z^2]\mathcal{K}(t, z; q) + \beta q^8 [z^0]\mathcal{K}(t, z; q). \quad (8)$$

Equivalently, in constant-term notation,

$$\text{CT}_z[(z^{-4} - \alpha q^6 z^{-2} - \beta q^8)S(tz, q)S(t/z, q)] = 0. \quad (9)$$

Proof. For any integer d ,

$$[z^d]\mathcal{K}(t, z; q) = \sum_{j \in \mathbb{Z}} W_{j+d} W_j t^{2j+d}.$$

Therefore

$$\begin{aligned}[z^4]\mathcal{K} &= \sum_{n \in \mathbb{Z}} W_{n+4} W_n t^{2n+4}, \\ [z^2]\mathcal{K} &= \sum_{n \in \mathbb{Z}} W_{n+3} W_{n+1} t^{2n+4}, \\ [z^0]\mathcal{K} &= \sum_{n \in \mathbb{Z}} W_{n+2}^2 t^{2n+4}.\end{aligned}$$

Substitution of (4) proves (8). The constant-term formulation follows from $[z^d]F(z) = \text{CT}_z(z^{-d}F(z))$. $\square$

Remark 1. *The identity is exact. No parity projection or truncation is required. The powers of t align automatically:*

$$(n) + (n + 4) = (n + 1) + (n + 3) = 2(n + 2) = 2n + 4.$$

Thus Somos-4 acts on one fixed sum diagonal and relates the three difference diagonals 4, 2, and 0.

4 Example: the classical Somos-4 sequence

For

$$\alpha = \beta = 1, \quad H_0 = H_1 = H_2 = H_3 = 1,$$

the bilateral extension begins

$$\dots, 8209, 1529, 314, 59, 23, 7, 3, 2, 1, 1, 1, 1, 2, 3, 7, 23, 59, 314, 1529, 8209, \dots$$

and satisfies

$$H_{-n} = H_{n+3}.$$

The transformed series begins

$$\begin{aligned} S(t, q) = & \dots + 7q^{12}t^{-3} + 3q^6t^{-2} + 2q^2t^{-1} + 1 + t + q^2t^2 + q^6t^3 \\ & + 2q^{12}t^4 + 3q^{20}t^5 + 7q^{30}t^6 + \dots \end{aligned}$$

The kernel identity specializes to

$$[z^4]\mathcal{K} = q^6[z^2]\mathcal{K} + q^8[z^0]\mathcal{K}. \quad (10)$$

5 Interpretation and research direction

The result gives a precise form to the proposed path

$$G(qx) \longrightarrow S(t, q) \longrightarrow \mathcal{K}(t, z; q).$$

The substitution $x \mapsto qx$ supplies the quadratic theta weight through Hankel scaling. The bilinear Somos recurrence then becomes a linear relation between three coefficient projections of a quadratic theta-like kernel.

The next questions are deliberately left open:

1. Can the projection relation (9) be converted into a closed functional equation using $z \mapsto q^k z$, a root-of-unity filter, or a suitable q -difference operator?
2. Does a distinguished normalization of S admit a factorization into Ramanujan general theta functions or their quotients?
3. Can the algebraic generating function for prescribed Somos-4 Hankel data be used to determine S without first solving the Somos orbit by elliptic uniformization?
4. What analytic region makes the bilateral series convergent for nontrivial Somos-4 growth? If ordinary convergence fails, which formal, regularized, or coefficientwise interpretation is most natural?
5. How does the construction change under a general gauge $H_n \mapsto AB^n C^{n^2} H_n$?

6 Status of the claim

The Hankel scaling (2), weighted recurrence (4), and diagonal kernel identity (8) are elementary exact consequences of the definitions. The possible identification of S with known theta or Ramanujan objects remains conjectural and is not asserted in this note.

References

- [1] T. Scheuerle, *Direct Generation of a Somos-4 Sequence from an Algebraic Generating Function*, arXiv:2609.15754, 2026.
- [2] F. Liu, Y. Wang, and Z. Zhang, *Hankel Transform and (α, β) Somos-4 Sequences*, arXiv:2608.08703, 2026.
- [3] For the standard notation $f(a, b)$ and the specialization $\varphi(q) = f(q, q) = \sum_{n \in \mathbb{Z}} q^{n^2}$, see standard accounts of Ramanujan's general theta function.